

Task 1

Domain: Software Project Management

PNG Tasks is a software project management system for organizing and monitoring software development projects. A project consists of work items arranged in a hierarchy, where task groups can contain individual tasks as well as other task groups. This allows large complex projects to be broken down into smaller manageable units.

Project managers can navigate the hierarchy in different ways to inspect and manage work. Individual tasks follow a lifecycle consisting of states such as Not Started, In Progress, Blocked, and Completed, with their behavior changing depending on their current state. Tasks can also acquire additional responsibilities at such times, such as priority handling, deadline monitoring, and notifications.

GoF Participants:

Iterator:

- ❖ **Iterator:** WorkItemIterator
- ❖ **ConcreteIterator:** DepthFirstIterator, TaskIterator
- ❖ **Aggregate:** WorkItem
- ❖ **ConcreteAggregate:** TaskGroup

Composite:

- ❖ **Component:** WorkItem
- ❖ **Leaf:** Task
- ❖ **Composite:** TaskGroup

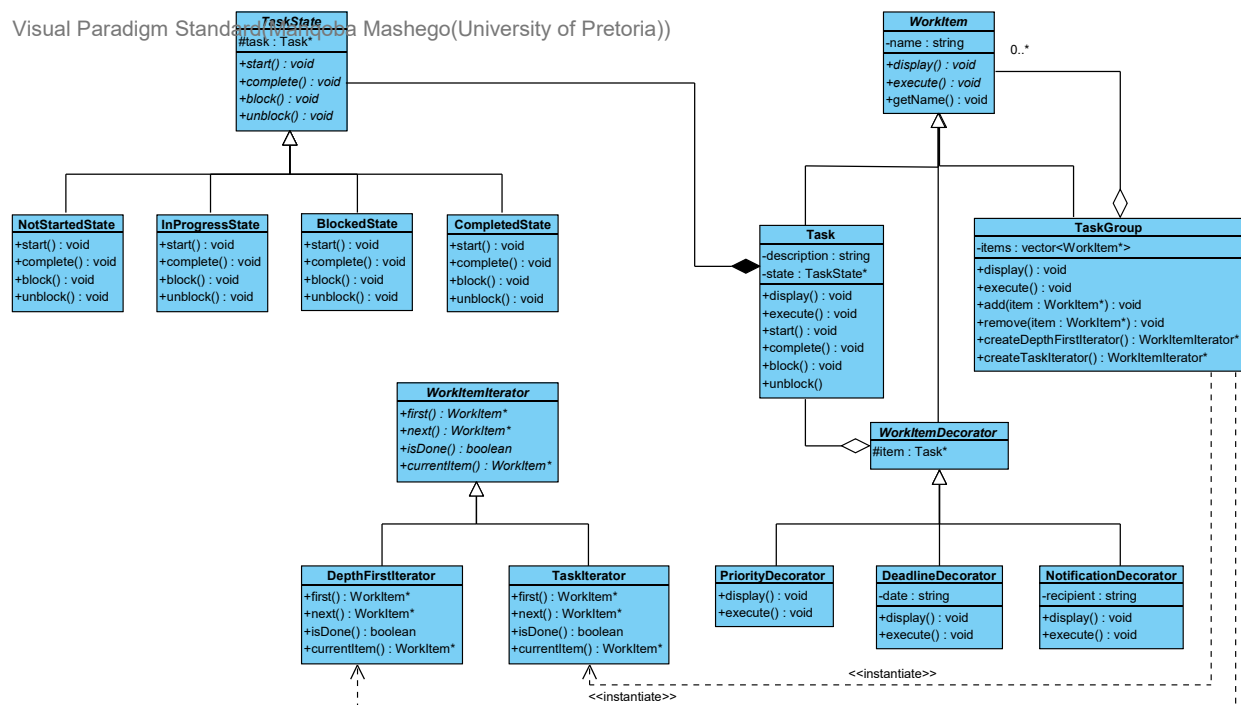
State:

- ❖ **State:** TaskState
- ❖ **ConcreteState:** NotStartedState, InProgressState, BlockedState, CompletedState
- ❖ **Context:** Task

Decorator:

- ❖ **Component:** WorkItem
- ❖ **ConcreteComponent:** Task
- ❖ **Decorator:** WorkItemDecorator
- ❖ **ConcreteDecorator:** PriorityDecorator, DeadlineDecorator, NotificationDecorator

Visual Paradigm Standard (Mashego Mashego (University of Pretoria))



Ownership policy:

TaskGroup owns the WorkItems that it contains and is responsible for deleting them. Since TaskGroup may contain other TaskGroup objects, destruction is recursive and destroys the whole hierarchy.

Task owns its current state. A Task is responsible for creating and deleting its current state. When a task transitions to another state, ownership of the previous state is released, and the task now owns the new state.

Decorators own the WorkItems they wrap. A decorator is responsible for deleting its WorkItem. This allows the decorators to be stacked, and each decorator owning the decorator or task it is wrapping.

Iterators do not own the WorkItems they traverse, they only maintain pointers to objects in the hierarchy.